

A PROJECT REPORT ON

SMART PARKING MANAGEMENT WITH NUMBER PLATE RECOGNITION

Domain: Computer Vision / Smart City

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CHAPTER I

INTRODUCTION

With the rapid increase in the number of cars and two-wheelers, finding a parking place has become a major problem near markets, temples, hospitals, bus stands, malls and offices. Drivers circle around looking for free space, which wastes fuel and adds to traffic congestion. Studies of urban traffic have found that a significant portion of traffic in busy areas is caused by vehicles searching for parking.

Most parking lots in India are managed manually. An attendant writes the vehicle number on a paper ticket at entry and calculates the fee at exit. This causes queues at the gate, errors in fee calculation, lost tickets and revenue leakage. Overstaying vehicles and vehicles parked illegally are not noticed, and there is no information for drivers about available spaces before they arrive.

This project, ParkEase, automates parking management using Automatic Number Plate Recognition (ANPR). Cameras at the entry and exit read number plates using a YOLOv8 detector and OCR, validate the Indian plate format, track free slots in real time, calculate fees automatically and generate digital receipts. Drivers can see the number of available spaces on a mobile web page before arriving.

LIMITATIONS

1. Drivers waste time and fuel searching for parking.
2. Manual ticketing causes queues and revenue leakage.
3. Overstay and illegal parking go unnoticed.
4. No real-time availability information for drivers.

1.1 History of Domain

Automatic number plate recognition was invented in 1976 at the Police Scientific Development Branch in the United Kingdom. Early ANPR systems used infrared cameras with classical image processing - edge detection, morphological operations and template matching - and were deployed for toll collection and law enforcement.

Deep learning based object detectors such as Faster R-CNN (2015) and YOLO (2016) greatly improved plate localisation under varied conditions, and CRNN-based OCR improved character recognition. Smart parking systems using ultrasonic sensors and mobile apps appeared in the 2010s, and ANPR-based ticketless parking is now used in many airports and malls.

1.2 Overview of the Project

At the entry gate, a camera captures the vehicle when it is detected by an IR sensor or motion detection. YOLOv8, fine-tuned on Indian number plates, locates the plate. The plate image is cropped, converted to grayscale, enhanced and passed to EasyOCR. The recognised text is cleaned and validated against the Indian registration format (for example TN 49 AB 1234, including BH series). The vehicle is assigned a free slot and the entry time is recorded.

At the exit gate, the plate is read again and matched with the active entry. Parking duration and fee are calculated using the tariff (first hour flat, then hourly, with separate rates for two-wheelers and cars), and the driver can pay by UPI QR code. Live slot availability is updated on the dashboard and the public mobile page. Alerts are raised for blacklisted vehicles and vehicles overstaying beyond the allowed time.

1.3 Motivation

A few seconds saved at every gate and a few minutes saved for every driver searching for parking add up to a big improvement in convenience and traffic flow.

- **Motivation:** To provide ticketless, queue-free and transparent parking with real-time availability.
- **Outcome:** Faster entry and exit, accurate revenue collection, reduced search traffic and better security.

1.4 Aim of the Project

The project is entitled as “**SMART PARKING MANAGEMENT WITH NUMBER PLATE RECOGNITION**”. The aim of this project is to develop a smart parking system that automatically recognises vehicle number plates at entry and exit, manages slot allocation in real time, calculates parking fees and alerts staff about blacklisted or overstaying vehicles.

1.5 Objectives

- To detect number plates using a YOLOv8 model trained on Indian plates.
- To read plate characters using OCR and validate the format.
- To allocate and track parking slots in real time.
- To compute fees and generate digital receipts.
- To provide live availability to drivers and alerts to staff.

1.6 Literature Review

Redmon et al. (2016): Proposed YOLO, which frames object detection as a single regression problem from image pixels to bounding boxes and class probabilities, achieving real-time speed.

Laroca et al. (2018): Presented a robust real-time ALPR system based on YOLO object detection for both plate detection and character recognition, and introduced the UFPR-ALPR dataset.

Shi, Bai and Yao (2017): Proposed the Convolutional Recurrent Neural Network (CRNN) for image-based sequence recognition, which is the basis of many modern OCR engines including EasyOCR.

Baek et al. (2019): Proposed CRAFT (Character Region Awareness for Text detection), which detects text regions by exploring each character and the affinity between characters.

Smith (2007): Described the architecture of the Tesseract OCR engine, which uses adaptive classification and line finding; widely used as an open-source OCR baseline.

Ministry of Road Transport and Highways: The Central Motor Vehicles Rules specify the format of Indian registration plates: state code, RTO code, series and number, and the BH series for vehicles of transferable employees.

CHAPTER II

SOFTWARE PROJECT PLAN

2.1 Existing System

In the existing system, an attendant manually writes or prints a token with the vehicle number and time at entry. At exit, the attendant calculates the duration and fee by looking at the token. Drivers must search physically for free slots. There is no record for audit and no alerts for overstaying vehicles.

DISADVANTAGES

- Queues at entry and exit gates.
- Human errors and revenue leakage.
- Lost tokens and disputes.
- No real-time slot information.
- No security alerts for suspicious vehicles.

2.2 Proposed System

The proposed ParkEase system uses cameras with YOLOv8-based plate detection and OCR to automate entry and exit. Slots are allocated automatically and availability is shown live. Fees are calculated by the system and paid through UPI. Blacklisted and overstaying vehicles trigger alerts, and all transactions are recorded for audit.

ADVANTAGES

- Ticketless, contactless entry and exit.
- Accurate, automatic fee calculation.
- Live availability for drivers.
- Alerts for blacklisted and overstaying vehicles.
- Complete audit trail and reports.

2.3 Project Schedule

The development of ParkEase follows the iterative waterfall model. Each phase produces a deliverable that is reviewed before the next phase begins, and feedback from testing is fed back into design and implementation. The planned schedule is shown below.

S.No	Phase	Duration
1	Requirement analysis and problem study	Week 1 - 2
2	Literature survey and existing system study	Week 3 - 4
3	Dataset collection / hardware procurement	Week 5 - 6
4	System design (DFD, UML, ER, table design)	Week 7 - 8
5	Implementation of modules	Week 9 - 12
6	Testing and model evaluation	Week 13 - 14
7	Documentation and final review	Week 15 - 16

CHAPTER III

CUSTOMER REQUIREMENT DETERMINATION

3.1 Functional Requirements

Functional requirements define the specific features and operations a system must perform to meet business and user needs. They describe what the system should do and how it interacts with users or other systems. They focus on the behaviour and functionality of the system and represent features that can be directly observed and tested in the final product. The functional requirements of the ParkEase system are listed below.

- **Vehicle detection:** Detect a vehicle at the gate.
- **Plate detection:** Locate the number plate using YOLOv8.
- **OCR:** Read plate characters.
- **Validation:** Validate Indian plate format and correct common OCR errors.
- **Slot allocation:** Assign a free slot and record entry time.
- **Live availability:** Update free slot count on dashboard and public page.
- **Fee and receipt:** Compute fee at exit and generate a digital receipt.
- **Alerts:** Alert on blacklisted or overstaying vehicles.

3.2 Non-functional Requirements

Non-functional requirements determine the performance standards and quality attributes of software, e.g. system usability, effectiveness, security, scalability, etc. While functional requirements determine what the system does, non-functional requirements describe HOW the system does it. If an application does not meet non-functional requirements, it continues to perform its basic functions; however, it will not be able to provide a good user experience. Therefore, non-functional requirements are as important as functional requirements for successful product adoption.

- **Performance:** Plate recognition within 1 second per vehicle.
- **Accuracy:** Plate read accuracy above 95% in daylight.
- **Reliability:** Manual override if recognition fails.

- **Availability:** 24x7 operation.
- **Security:** Role-based access for operators and admin.
- **Usability:** Public page loads quickly on mobile.

3.3 User Requirements

- Drivers see available slots before arriving
- Quick entry and exit without tickets
- Fee shown clearly with digital receipt
- Operator can override if plate is unreadable
- Owner can view revenue reports

3.4 System Requirements

- IP/USB cameras at entry and exit
- Server with Python, YOLOv8 and EasyOCR
- MySQL database
- Flask API and React dashboard
- UPI payment QR

CHAPTER IV

SOFTWARE REQUIREMENT SPECIFICATION

4.1 Hardware Specification

System : Intel i5 Processor with NVIDIA GPU (recommended)

Hard Disk : 512 GB

RAM : 16 GB

Cameras : 2 x 1080p IP cameras with IR

Sensors : IR vehicle detection sensor at gates (optional)

Gate : Boom barrier with relay control (optional)

4.2 Software Specification

Operating System : Ubuntu 22.04 / Windows 10

Coding Language : Python 3.10, JavaScript

Detection : YOLOv8 (Ultralytics)

OCR : EasyOCR / Tesseract

Back End : Flask

Front End : React JS

Database : MySQL

CHAPTER V

SYSTEM ANALYSIS

5.1 Feasibility Study

A feasibility study is carried out to find whether the proposed ParkEase system is worth developing. Three key considerations are involved in the feasibility analysis: technical, economic and operational feasibility.

TECHNICAL FEASIBILITY

YOLOv8 and EasyOCR are open source and achieve real-time performance on a mid-range GPU. Public Indian number plate datasets are available for fine-tuning. The system is technically feasible.

ECONOMIC FEASIBILITY

Two cameras and a computer cost much less than the revenue lost through leakage in a busy parking lot within months. All software is open source. It is economically feasible.

OPERATIONAL FEASIBILITY

Operators only monitor the dashboard and handle exceptions. Drivers do not need to install anything. The system is operationally feasible.

5.2 About the Software

PYTHON

Python is a high-level, interpreted, general-purpose programming language created by Guido van Rossum and first released in 1991. Its design philosophy emphasises code readability through significant indentation, and its simple syntax allows developers to express concepts in fewer lines of code than languages such as C++ or Java.

Python supports multiple programming paradigms including procedural, object-oriented and functional programming. It has a very large standard library and an

ecosystem of third-party packages available through the Python Package Index (PyPI), which makes it the most popular language for data science, machine learning, automation and back-end web development.

- Easy to learn, read and maintain
- Huge library support for AI, ML and data processing (NumPy, Pandas, scikit-learn)
- Platform independent - runs on Windows, Linux and macOS
- Strong community support and documentation

YOLOv8

YOLO (You Only Look Once) is a family of real-time object detection models. YOLOv8, released by Ultralytics, detects objects in a single forward pass of a convolutional neural network and returns bounding boxes and class probabilities.

YOLOv8 can be fine-tuned on a custom dataset, such as number plates, with only a few hundred labelled images, and runs at real-time speed even on a normal laptop GPU.

- Single-stage, very fast detector
- Easy training with the Ultralytics Python API
- Export to ONNX and TensorRT

EASYOCR / TESSERACT OCR

Optical Character Recognition (OCR) converts images of typed or printed text into machine-encoded text. Tesseract is an open-source OCR engine sponsored by Google, and EasyOCR is a deep-learning based OCR library supporting more than 80 languages.

EasyOCR uses a CRAFT text detector and a CRNN recogniser, and performs well on the irregular fonts and lighting found on vehicle number plates.

OPENCV

OpenCV (Open Source Computer Vision Library) is an open-source computer vision and machine learning software library containing more than 2500 optimised

algorithms. It was originally developed by Intel and is now maintained by the OpenCV foundation.

OpenCV is used to capture video from cameras, resize and convert images, detect edges and contours, apply filters and run deep neural network models through its dnn module.

- Real-time image and video processing
- Python, C++ and Java interfaces
- Hardware acceleration support

FLASK

Flask is a lightweight WSGI web application framework written in Python. It is classified as a micro-framework because it does not require particular tools or libraries; it has no database abstraction layer or form validation built in, but extensions can add these features as needed.

Flask provides URL routing, request handling, Jinja2 templating and a development server. Because it is small and simple, it is ideal for exposing a trained machine learning model as a REST API or building a small web application quickly.

- Minimal and easy to learn
- Jinja2 templates for HTML pages
- Large set of extensions (Flask-Login, Flask-SQLAlchemy, Flask-CORS)

REACT JS

React is an open-source JavaScript library for building user interfaces, maintained by Meta and a community of developers. It lets developers build reusable UI components and manages the view layer of web applications efficiently using a virtual DOM.

React applications are composed of components that hold their own state and re-render automatically when data changes. With libraries such as React Router and Axios, a single-page application can communicate with REST APIs and give the user a fast, app-like experience.

- Component based, reusable UI
- Virtual DOM for fast rendering
- Large ecosystem of UI libraries

MYSQL

MySQL is an open-source relational database management system (RDBMS) based on Structured Query Language. It is one of the most widely used databases in web applications and is known for its speed, reliability and ease of use.

MySQL stores data in tables made of rows and columns, enforces relationships using primary and foreign keys, and supports transactions through the InnoDB storage engine.

- Fast and reliable
- Easy to install and manage with tools like MySQL Workbench
- Supported by all major programming languages

CHAPTER VI

SYSTEM DESIGN

System design is the process of defining the architecture, components, modules, interfaces and data of a system to satisfy the specified requirements. The design of ParkEase is presented using UML use case diagram, system architecture, data flow diagrams, ER diagram and table design.

6.1 UML Diagram

USE CASE DIAGRAM

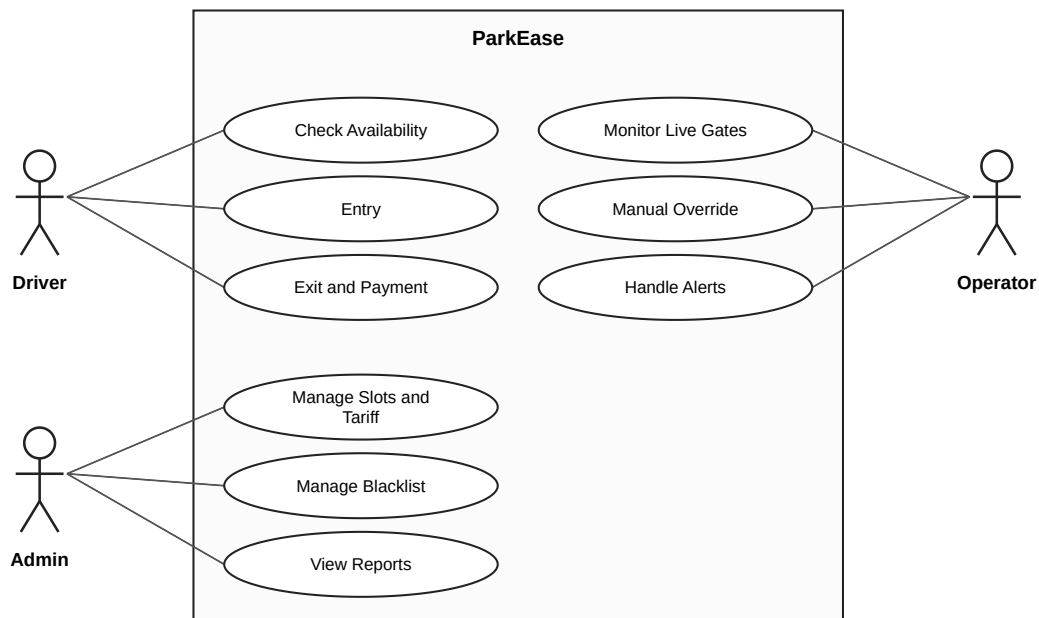


Fig 1: Use case diagram of ParkEase

6.2 System Architecture

ParkEase has an edge vision layer at the gates, a central application server and a database, with dashboards for operators and a public availability page.

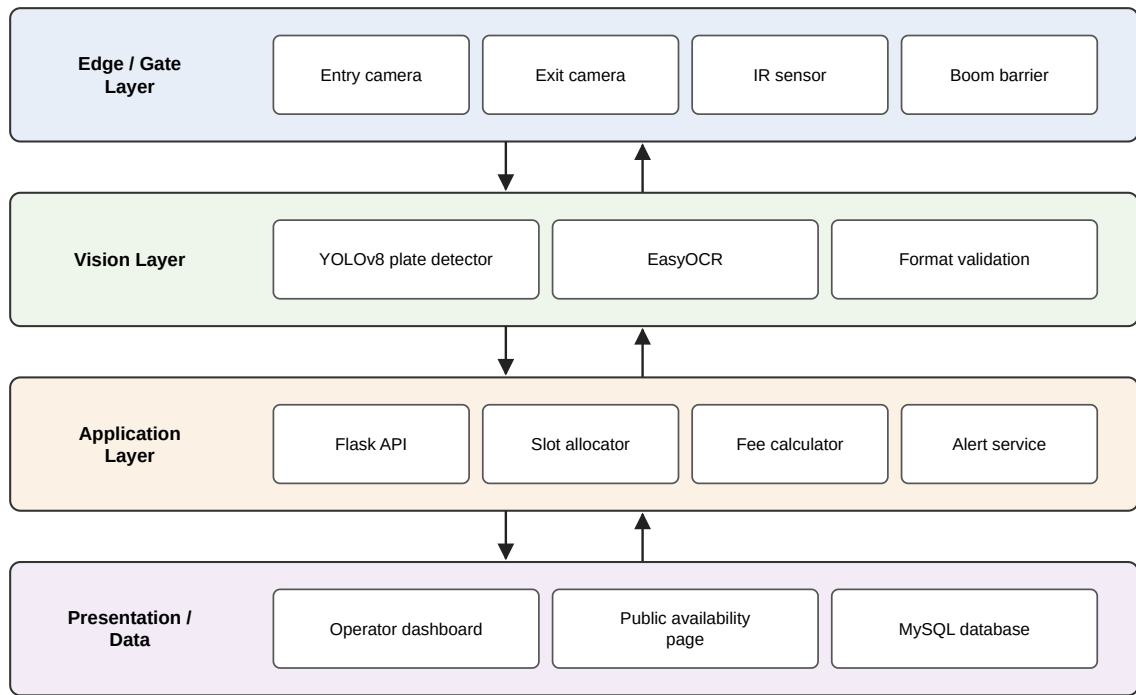


Fig 2: System architecture

6.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the flow of data through an information system. The Level 0 DFD (context diagram) shows the whole system as a single process with its external entities, and the Level 1 DFD breaks it into its major processes and data stores.

LEVEL 0 DFD (CONTEXT DIAGRAM)

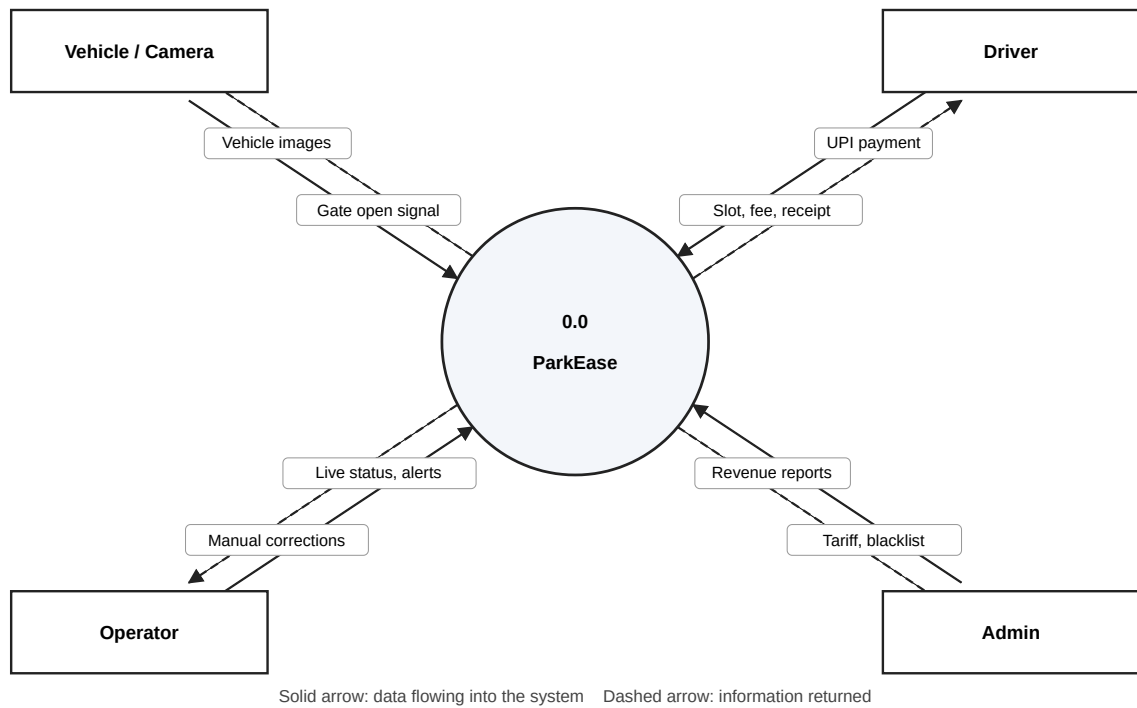


Fig 3: Level 0 data flow diagram

LEVEL 1 DFD

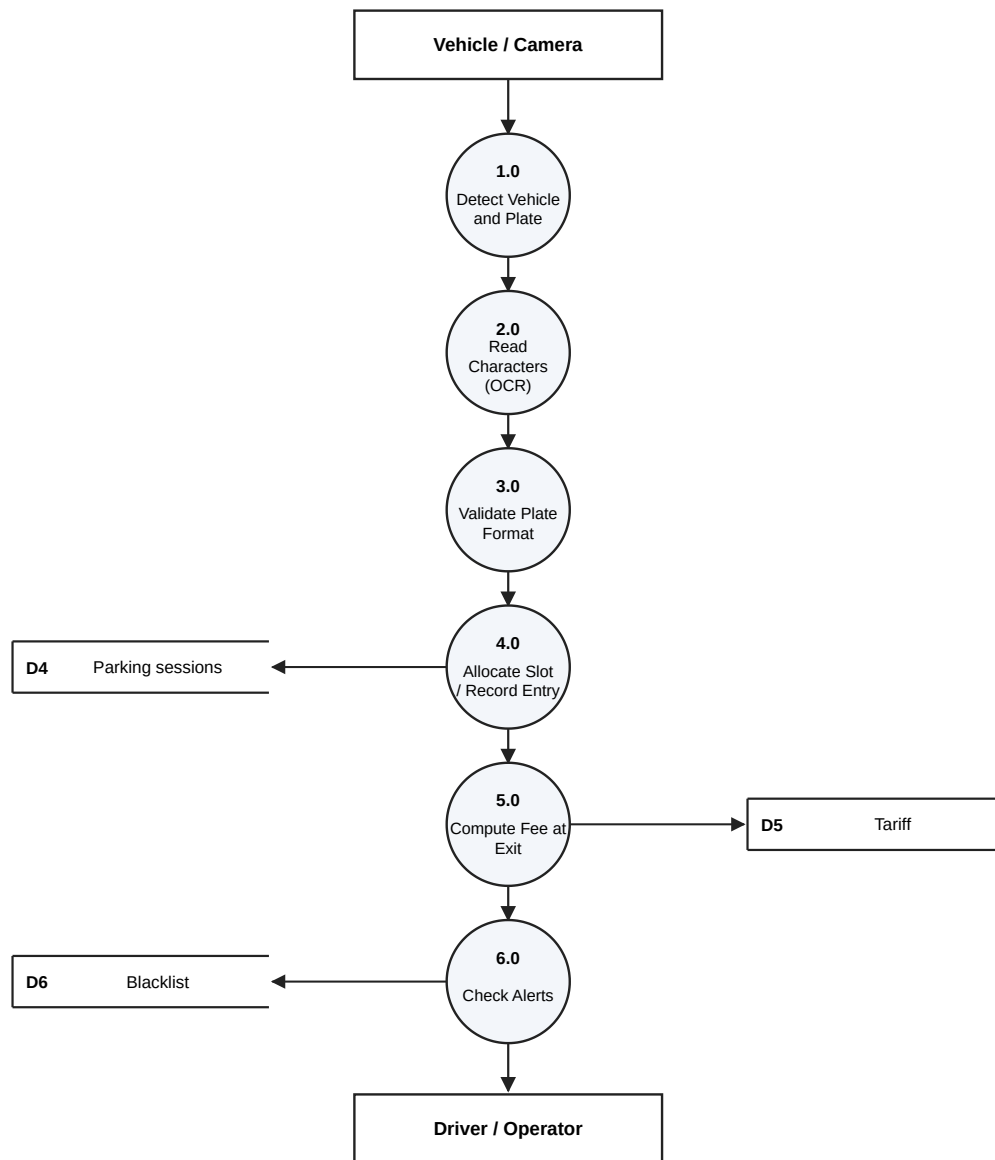


Fig 4: Level 1 data flow diagram

6.4 ER Diagram

The Entity Relationship (ER) diagram shows the entities stored in the database, their key attributes and the relationships between them. 1:N indicates a one-to-many relationship.

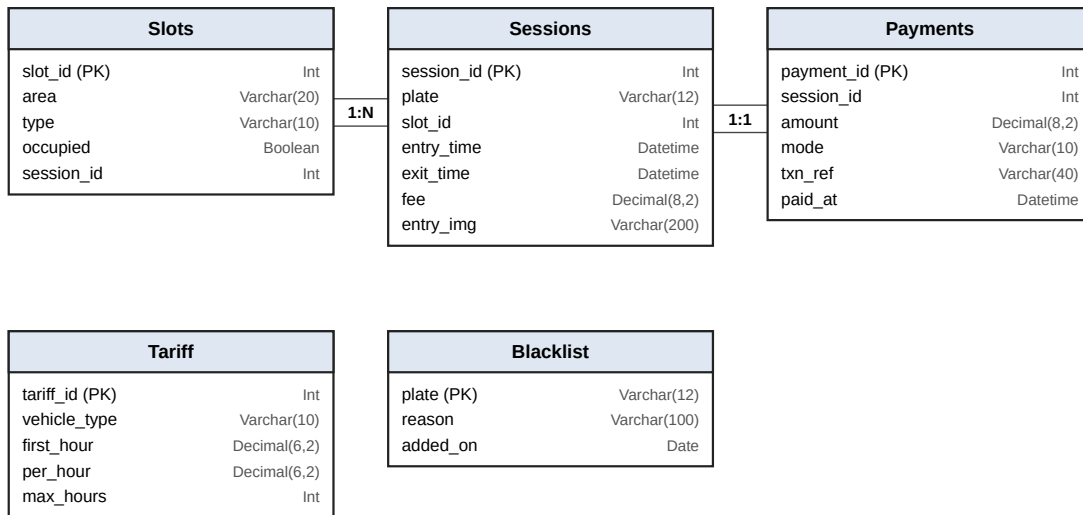


Fig 5: Entity relationship diagram

6.5 Table Design

SLOTS

FIELD NAME	DATA TYPE	DESCRIPTION
slot_id	Int	Primary key
area	Varchar(20)	Area / level
type	Varchar(10)	car / bike
occupied	Boolean	Status
session_id	Int	Current session

SESSIONS

FIELD NAME	DATA TYPE	DESCRIPTION
session_id	Int	Primary key
plate	Varchar(12)	Vehicle number
slot_id	Int	Foreign key to Slots
entry_time	Datetime	Entry
exit_time	Datetime	Exit
fee	Decimal(8,2)	Fee
entry_img	Varchar(200)	Entry snapshot

PAYMENTS

FIELD NAME	DATA TYPE	DESCRIPTION
payment_id	Int	Primary key
session_id	Int	Foreign key to Sessions
amount	Decimal(8,2)	Amount
mode	Varchar(10)	UPI / cash
txn_ref	Varchar(40)	UPI reference
paid_at	Datetime	Time

TARIFF

FIELD NAME	DATA TYPE	DESCRIPTION
tariff_id	Int	Primary key
vehicle_type	Varchar(10)	car / bike
first_hour	Decimal(6,2)	First hour rate
per_hour	Decimal(6,2)	Additional hour rate
max_hours	Int	Allowed hours

BLACKLIST

FIELD NAME	DATA TYPE	DESCRIPTION
plate	Varchar(12)	Primary key
reason	Varchar(100)	Reason
added_on	Date	Date

6.6 Algorithm Design

The ANPR and parking algorithm runs at both gates; the exit gate additionally computes the fee.

1. **Detect vehicle at gate** – IR sensor trigger or frame difference detects a vehicle; the best frame from 5 captured frames is chosen.

2. **Locate number plate with YOLO** – YOLOv8n fine-tuned on Indian plates returns the plate bounding box with confidence.
3. **Read characters with OCR** – The crop is converted to grayscale, resized, contrast enhanced (CLAHE) and passed to EasyOCR with an alphanumeric allow-list.
4. **Validate Indian plate format with regex** – Pattern $^{[A-Z]\{2\}[0-9]\{1,2\}[A-Z]\{0,3\}[0-9]\{4\}}\$$ or BH series; position-based correction (O->0, I->1, B->8) is applied.
5. **Record entry time and allocate slot** – The nearest free slot of the correct vehicle type is allocated.
6. **Update live availability** – Slot counts are pushed to dashboards via WebSocket.
7. **Compute fee at exit and generate receipt** – $\text{fee} = \text{first_hour} + \text{ceil}(\text{hours} - 1) \times \text{per_hour}$; a receipt with entry/exit times is generated.
8. **Alert on blacklisted or overstayed vehicles** – Plates are checked against the blacklist at entry; a periodic job flags sessions longer than max_hours.

PSEUDO CODE

```

ON VEHICLE_AT_GATE(frames, gate):
  best <- argmax over frames of YOLO(frame).plate_conf
  plate_img <- CROP(best); txt <- OCR(ENHANCE(plate_img))
  plate <- CORRECT_AND_VALIDATE(txt)
  IF plate invalid THEN ASK_OPERATOR(plate_img); RETURN
  IF gate = ENTRY:
    IF plate IN blacklist THEN ALERT("Blacklisted", plate)
    slot <- FREE_SLOT(type); CREATE_SESSION(plate, slot, now)
  ELSE:
    s <- ACTIVE_SESSION(plate); h <- HOURS(now - s.entry)
    fee <- tariff.first + CEIL(MAX(h-1, 0)) * tariff.per_hour
    SHOW_UPI_QR(fee); ON PAID: CLOSE_SESSION(s); FREE(s.slot); OPEN_GATE()

```

6.7 Modules

The modules are grouped by the Driver, Operator and Admin users.

MODULES

DRIVER

1. CHECK AVAILABILITY

2. ENTRY

3. EXIT AND PAYMENT

OPERATOR

1. MONITOR LIVE GATES

2. MANUAL OVERRIDE

3. HANDLE ALERTS

ADMIN

1. MANAGE SLOTS AND TARIFF

2. MANAGE BLACKLIST

3. VIEW REPORTS

DRIVER

CHECK AVAILABILITY

Drivers open the public page to see free slots for cars and two-wheelers in each parking area.

ENTRY

The plate is read automatically at the gate, a slot is assigned and displayed on the gate screen.

EXIT AND PAYMENT

At exit, the fee is displayed with a UPI QR code; after payment, a digital receipt is sent and the gate opens.

OPERATOR

MONITOR LIVE GATES

Operators see live camera feeds with recognised plates and confidence.

MANUAL OVERRIDE

If the plate cannot be read (damaged or dirty plate), the operator types the number manually.

HANDLE ALERTS

Operators receive alerts for blacklisted and overstaying vehicles.

ADMIN**MANAGE SLOTS AND TARIFF**

Admin configures parking areas, slot counts and tariff rules.

MANAGE BLACKLIST

Admin adds or removes vehicles from the blacklist.

VIEW REPORTS

Admin views daily revenue, occupancy and peak hours.

CHAPTER VII

IMPLEMENTATION AND TESTING

7.1 Introduction

ParkEase was implemented in Python. The YOLOv8n model was fine-tuned using the Ultralytics library on about 1,500 labelled images of Indian number plates for 60 epochs. EasyOCR performs character recognition. The Flask server handles gate events, and a React dashboard displays live slots via Socket.IO.

7.2 Implementation

The system was tested in a college parking area with two cameras for one week, covering daytime, evening and night conditions with IR illumination. Common OCR confusions were reduced by position-aware correction based on the plate format.

Testing is a process of running software with the intention of finding errors. The software which has been developed has to be tested to prove its validity. Testing is considered to be the least creative phase of the whole cycle of system design; in the real sense it is the phase which helps to bring out the creativity of the other phases. Testing a program is the first step in the debugging process.

A good testing suite is vital because all possible execution paths inside the code must be tested in order to have any hope that it is bug free. Testing of the system is done to ensure the integrity of the system. The aim of testing is to prove that the developed system addresses the predefined processing requirements and will perform reliably and efficiently during operation.

PURPOSES OF TESTING

- To affirm the quality of the project.
- To find and eliminate any errors from previous stages.
- To validate the software against its requirements.
- To ensure the reliability of the system.

UNIT TESTING

In the lines of this strategy all the individual functions and modules were put to the test independently. By following this strategy all the errors in coding were identified and corrected. This method was applied in combination with white box and black box testing techniques to find the errors in each module.

INTEGRATION TESTING

This testing strategy has two different approaches, namely the top-down approach, in which integration is carried out from the top-level module to the bottom, and the bottom-up approach, in which integration is carried out from the low-level modules to the top. The modules were tested using the bottom-up approach by introducing stubs for the top-level functions. This test was used to identify errors in the interfaces and in passing parameters between functions.

VALIDATION TESTING

The main aim of this testing is to verify that the software system does what it was designed for. Alpha testing and beta testing were conducted to ensure the validity of the system. Alpha testing was done at the developer's site and beta testing was carried out by sample end users.

SYSTEM TESTING

The idea of applying this strategy was to put the system through a series of different tests whose primary purpose is to fully exercise the computer-based system. Although each test has a different purpose, all work to verify that all system elements have been properly integrated and perform their allocated functions. Security testing and performance testing were also conducted.

ACCEPTANCE TESTING

Acceptance testing involves planning and execution of functional tests, performance tests and stress tests to verify that the implemented system satisfies its

requirements. Typically, the acceptance test incorporates test cases developed during unit testing and integration testing.

7.3 Test Cases

Test ID	Test Description	Input	Expected Output	Result
TC01	Plate detection	Car image at gate	Plate bounding box found	Pass
TC02	OCR reading	Clear plate TN49AB1234	Text 'TN49AB1234'	Pass
TC03	OCR correction	Read as 'TN49AB12E4' / 'TN4OAB1234'	Corrected to valid format or sent to operator	Pass
TC04	Slot allocation	Car entry	Free car slot assigned	Pass
TC05	Fee calculation	Stay 2 h 20 min, car	Rs. 20 + 2 x 15 = Rs. 50	Pass
TC06	Blacklisted vehicle	Plate in blacklist	Alert to operator	Pass
TC07	Overstay	Bike parked 13 h (max 12)	Overstay alert	Pass
TC08	Parking full	All car slots occupied	Display 'FULL' at gate and page	Pass

PERFORMANCE EVALUATION

Field test results (1,240 vehicle passages):

Condition	Plate detection	Full plate read accuracy	Avg time
Daylight	99.1%	96.4%	0.42 s
Evening	98.0%	93.2%	0.45 s
Night (IR)	96.5%	90.8%	0.47 s
Overall	98.2%	94.1%	0.44 s

7.4 Source Code

anpr.py

```
import re, cv2, numpy as np, easyocr
from ultralytics import YOLO
```

```

detector = YOLO("models/plate_yolov8n.pt")
reader = easyocr.Reader(["en"], gpu=True)
PATTERN = re.compile(r"^[A-Z]{2}[0-9]{1,2}[A-Z]{0,3}[0-9]{4}$")
BH = re.compile(r"^[0-9]{2}BH[0-9]{4}[A-Z]{1,2}$")
TO_DIGIT = {"0": "0", "D": "0", "I": "1", "L": "1", "Z": "2", "S": "5", "B": "8", "G": "6"}
TO_ALPHA = {v: k for k, v in {"0": "0", "I": "1", "Z": "2", "S": "5", "B": "8", "G": "6"}.items()}

def enhance(img):
    g = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    g = cv2.resize(g, None, fx=2, fy=2, interpolation=cv2.INTER_CUBIC)
    return cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8)).apply(g)

def correct(text):
    t = re.sub(r"^[A-Z0-9]", "", text.upper())
    if len(t) < 8: return t
    chars = list(t)
    for i in (0, 1):
        # state code must be letters
        chars[i] = TO_ALPHA.get(chars[i], chars[i])
    for i in (2, 3):
        # RTO code digits
        chars[i] = TO_DIGIT.get(chars[i], chars[i])
    for i in range(len(chars) - 4, len(chars)):
        # last four are digits
        chars[i] = TO_DIGIT.get(chars[i], chars[i])
    return "".join(chars)

def read_plate(frames):
    best, best_conf = None, 0
    for f in frames:
        r = detector(f, verbose=False)[0]
        for box in r.bboxes:
            if float(box.conf) > best_conf:
                x1, y1, x2, y2 = map(int, box.xyxy[0])
                best, best_conf = f[y1:y2, x1:x2], float(box.conf)
    if best is None:
        return None, 0.0
    results = reader.readtext(enhance(best),
allowlist="ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789", detail=1)
    text = "".join(t for _, t, _ in sorted(results, key=lambda r: (r[0][0][1] //
30, r[0][0][0])))
    plate = correct(text)
    ok = bool(PATTERN.match(plate) or BH.match(plate))
    conf = np.mean([c for _, c in results]) if results else 0
    return (plate if ok else None), float(conf) * best_conf

```

gate_service.py

```

import math, datetime as dt, mysql.connector
from flask import Flask, request, jsonify
from flask_socketio import SocketIO
from anpr import read_plate
import cv2, numpy as np

app = Flask(__name__); sio = SocketIO(app, cors_allowed_origins="*")
db = mysql.connector.connect(user="park", password="secret",
database="parkease", autocommit=True)

def q(sql, *a):
    cur = db.cursor(dictionary=True); cur.execute(sql, a)
    return cur.fetchall() if cur.with_rows else None

```

```

def frames_from_request():
    return [cv2.imdecode(np.frombuffer(f.read(), np.uint8), 1) for f in
request.files.getlist("frames")]

@app.post("/gate/entry")
def entry():
    plate, conf = read_plate(frames_from_request())
    if not plate:
        return jsonify(status="manual", message="Plate unreadable - operator
input needed")
    vtype = request.form.get("type", "car")
    if q("SELECT 1 FROM blacklist WHERE plate=%s", plate):
        sio.emit("alert", {"type": "blacklisted", "plate": plate})
    slot = q("SELECT slot_id FROM slots WHERE type=%s AND occupied=0 ORDER BY
slot_id LIMIT 1", vtype)
    if not slot:
        return jsonify(status="full")
    sid = slot[0]["slot_id"]
    q("INSERT INTO sessions(plate, slot_id, entry_time) VALUES (%s,%s,NOW())",
plate, sid)
    q("UPDATE slots SET occupied=1 WHERE slot_id=%s", sid)
    sio.emit("slots", q("SELECT type, SUM(occupied=0) AS free FROM slots GROUP
BY type"))
    return jsonify(status="ok", plate=plate, slot=sid, confidence=round(conf,
2))

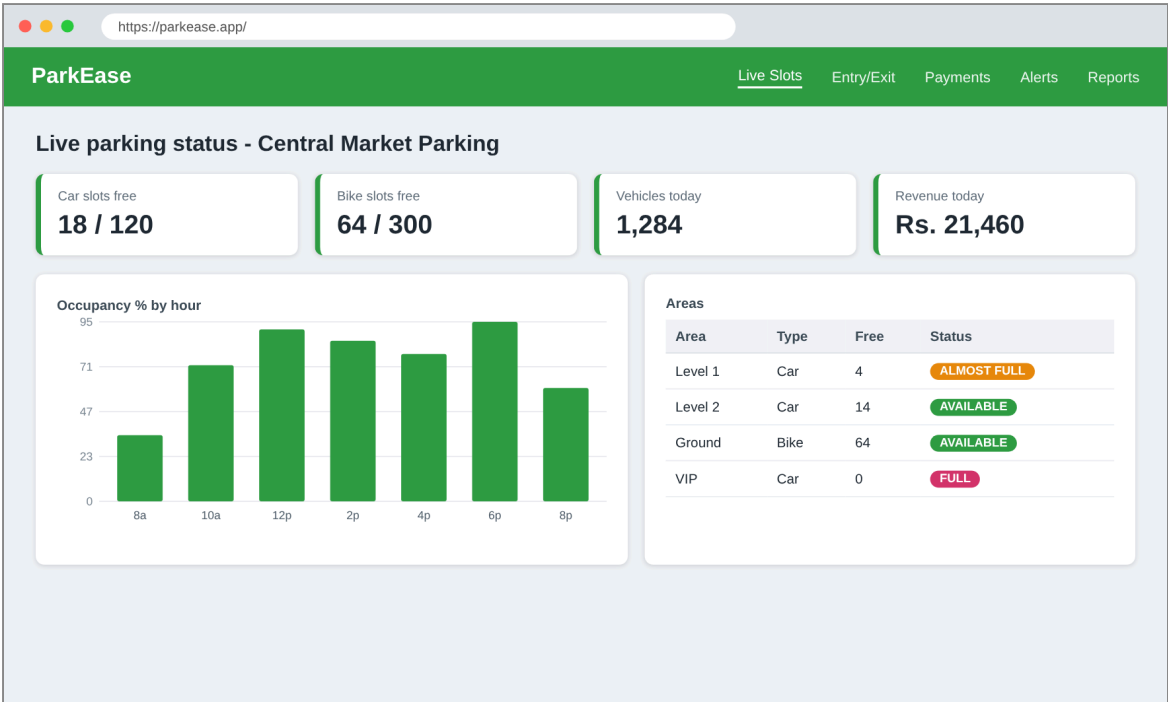
@app.post("/gate/exit")
def exit_gate():
    plate, _ = read_plate(frames_from_request())
    s = q("""SELECT s.session_id, s.slot_id, s.entry_time, t.first_hour,
t.per_hour FROM sessions s
        JOIN slots sl ON sl.slot_id=s.slot_id JOIN tariff t ON
t.vehicle_type=sl.type
        WHERE s.plate=%s AND s.exit_time IS NULL""", plate)
    if not s:
        return jsonify(status="not_found", plate=plate)
    s = s[0]
    hours = (dt.datetime.now() - s["entry_time"]).total_seconds() / 3600
    fee = float(s["first_hour"]) + math.ceil(max(hours - 1, 0)) *
float(s["per_hour"])
    q("UPDATE sessions SET exit_time=NOW(), fee=%s WHERE session_id=%s", fee,
s["session_id"])
    q("UPDATE slots SET occupied=0 WHERE slot_id=%s", s["slot_id"])
    return jsonify(status="ok", plate=plate, hours=round(hours, 2), fee=fee,
        upi=f"upi://pay?pa=parkease@upi&am={fee:.2f}&tn={plate}")

```

7.5 Screen Layout

SCREENSHOTS:

LIVE SLOT AVAILABILITY



EXIT AND PAYMENT

https://parkease.app/

ParkEase

Live SlotsEntry/ExitPaymentsAlertsReports

Exit Gate - Payment

Fee: Rs. 50 (2 h 20 min)

Vehicle	TN 49 AB 1234
Entry	10:42
Exit	13:02
Tariff	Rs. 20 first hour + Rs. 15 per extra hour
Payment	Scan UPI QR - parkease@upi

Receipt will be sent by SMS after payment.

ALERTS

https://parkease.app/

ParkEase

Live SlotsEntry/ExitPaymentsAlertsReports

Alerts

Time	Plate	Type	Details	Status
14-10 11:05	TN 07 CQ 4471	BLACKLIST	Reported stolen	Police informed
14-10 09:30	TN 49 Z 9921	OVERSTAY	Parked 13 h (max 12)	Notice issued
13-10 18:12	UNREADABLE	MANUAL	Damaged plate	Entered by operator

CHAPTER VIII

CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

The Smart Parking Management system with Number Plate Recognition automates entry, exit and fee collection using YOLOv8 and OCR, achieving 94.1% full-plate read accuracy with an average recognition time of 0.44 seconds. Real-time slot availability reduces the time drivers spend searching for parking, and automatic fee calculation eliminates errors and revenue leakage.

Alerts for blacklisted and overstaying vehicles improve security. The system provides a practical, low-cost smart city solution for markets, hospitals, temples and malls.

8.2 Future Enhancement

- Per-slot occupancy detection with overhead cameras or ultrasonic sensors.
- Advance slot booking through the mobile app.
- FASTag-based automatic payment.
- Integration with city-wide parking guidance boards.
- Electric vehicle charging slot management.

CHAPTER IX

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